

Isometric unit spheres force equal finite dimensions

MathlibAnnex.Sphere.finrank_eq

theorem

Compares Hausdorff dimensions in both directions using the surjective Lipschitz radial maps.

Statement

Let X, Y be finite-dimensional real normed spaces and let $\Delta : S_X \rightarrow S_Y$ be a bijective isometry for the chord metrics inherited from their ambient norms. Here $S_X = \{u \in X : \|u\| = 1\}$ and $S_Y = \{v \in Y : \|v\| = 1\}$ are the unit spheres. Then

$$\dim_{\mathbb{R}} X = \dim_{\mathbb{R}} Y.$$

Assumptions

Both X and Y are finite-dimensional in this exact theorem. Each carries a real normed-space structure and Δ is a sphere isometry equivalence. If only one space is known to be finite-dimensional, the separate transfer theorem first supplies the other hypothesis. No positive-dimension assumption is needed.

Conclusion

The natural-number dimensions of the two real vector spaces agree. The intermediate invariant is the Hausdorff dimension of the entire ambient metric space, which is an extended nonnegative real number; its identification with linear dimension is a separate finite-dimensional theorem.

The proof compares the forward and inverse radial maps. It neither uses an unproved invariance of topological dimension nor identifies either radial map with a linear map.

Proof route

Write $\dim_H(X)$ for the Hausdorff dimension of the whole metric space X . The map $R_\Delta : X \rightarrow Y$ is 3-Lipschitz, and the inverse identities show it is surjective. The Lipschitz range theorem therefore gives

$$\dim_H(Y) = \dim_H(R_\Delta(X)) \leq \dim_H(X).$$

For the inverse sphere isometry, $R_{\Delta^{-1}} : Y \rightarrow X$ is likewise surjective and 3-Lipschitz, giving

$$\dim_H(X) = \dim_H(R_{\Delta^{-1}}(Y)) \leq \dim_H(Y).$$

Thus the Hausdorff dimensions are equal. For each finite-dimensional real normed space the Hausdorff dimension of the whole space equals its real-linear dimension, viewed in $[0, \infty]$. Applying that theorem to both spaces and using injectivity of the natural-number cast gives the asserted equality of linear dimensions.

▼ Proof steps

1. Use the forward radial map with the Lipschitz range bound, then replace its range by all of Y using surjectivity. This proves the inequality in the direction from Y to X .
2. Repeat with the inverse sphere isometry to obtain the opposite inequality. Hence the Hausdorff dimensions agree. For finite-dimensional real normed spaces these equal the real-linear dimensions, including in dimension zero.

Main citations

- Exact declaration and its source — `MathlibAnnex.Sphere.finrank_eq`
- The forward Lipschitz bound — `MathlibAnnex.Sphere.lipschitzWith_radialExtension`
- Surjectivity of the radial extension — `MathlibAnnex.Sphere.radialExtension_surjective`
- The inverse Lipschitz bound — `MathlibAnnex.Sphere.lipschitzWith_radialExtension_symm`
- The two Hausdorff-dimension comparisons in the proof — `MathlibAnnex.Sphere.dimH_univ_eq`
- Hausdorff dimension of a Lipschitz range — `LipschitzWith.dimH_range_le`
- Hausdorff dimension equals finite real dimension — `Real.dimH_univ_eq_finrank`
- Finite-dimensionality from one side — `MathlibAnnex.Sphere.finiteDimensional_codomain`

Lean source signature (exact)

```
theorem finrank_eq
  [FiniteDimensional ℝ X] [FiniteDimensional ℝ Y]
  (e : Metric.sphere (0 : X) 1 ≈i Metric.sphere (0 : Y) 1) :
  Module.finrank ℝ X = Module.finrank ℝ Y
```

In the source	Mathematical meaning
<code>(e : Metric.sphere (0 : X) 1 ≈_i Metric.sphere (0 : Y) 1)</code>	The same bijective chord-distance isometry $\Delta : S_X \rightarrow S_Y$, where $S_X = \{x : \ x\ _X = 1\}$ and $S_Y = \{y : \ y\ _Y = 1\}$. The surrounding binders make X, Y real normed spaces.
<code>[FiniteDimensional ℝ X] [FiniteDimensional ℝ Y]</code>	Both X and Y are finite-dimensional inputs in this theorem.

In the source	Mathematical meaning
<code>Module.finrank ℝ X; Module.finrank ℝ Y</code>	The natural-number dimensions $\dim_{\mathbb{R}} X$ and $\dim_{\mathbb{R}} Y$.
<code>Module.finrank ℝ X = Module.finrank ℝ Y</code>	The conclusion $\dim_{\mathbb{R}} X = \dim_{\mathbb{R}} Y$. The extended nonnegative Hausdorff dimensions in the proof are a separate intermediate invariant.

▼ Exact surrounding binder context (separate excerpt)

```

namespace MathlibAnnex
namespace Sphere

universe u v

variable {X : Type u} {Y : Type v}
  [NormedAddCommGroup X] [NormedSpace ℝ X]
  [NormedAddCommGroup Y] [NormedSpace ℝ Y]

```

▼ Exact content identity

Declaration: `MathlibAnnex.Sphere.finrank_eq`

Accepted content SHA-256: `2c49c2c27a365647a173f2444ec55264f482baf0c87529e776c4b4361b783ff8`

Accepted source guide SHA-256: `0ac27177223449fcb58a31f86aba75b2be5119a0b4d4097a76a3d18258a8885f`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`